

LB-AO Quantitative Property Schema

Technical rationale and gluteus maximus reference-muscle example

Purpose. Record why the project treats a quantitative parameter as more than a number. The schema keeps values together with evidence, uncertainty, applicability, transformations, coordinate/model context, assumptions, and validation status so the model can be inspected, reproduced, challenged, and later personalized.

Guiding principle: A model may begin simple, but its simplifications and evidence must remain visible.

1. The problem the schema solves

Musculoskeletal quantities are context dependent. Optimal fiber length, physiological cross-sectional area (PCSA), maximum isometric force, joint-center position, moment arm, and segment inertia can vary with measurement method, population, coordinate convention, scaling procedure, or model formulation. Storing only a scalar value hides those dependencies.

LB-QPS therefore separates the anatomical entity from observations about that entity and separates experimental observations from transformations used to create computational parameters. Multiple defensible observations may coexist rather than being silently overwritten.

2. Rules adopted for quantitative data

Schema first; values second. Define the property's meaning, unit, role, and required context before populating it.

No unsupported filling. Unknown values remain explicitly unresolved instead of receiving convenient estimates.

Claim-specific evidence. A source supports named claims, not every field attached to an entity.

Preserve derivations. Calculated/model-derived values retain upstream observations, equations, conversions, and assumptions.

Applicability travels with the value. Population, sample size, model, subject, and other relevant context remain attached.

Uncertainty is retained. Reported spread or uncertainty is not discarded during unit conversion or model translation.

Coordinate context is required when needed. Angles, positions, moments, forces, and moment arms require frames/conventions.

Assumptions remain assumptions. For example, bilateral symmetry is a reversible baseline policy, not anatomical truth.

Backward compatibility is tested. Stable IDs and accumulated tests protect earlier ontology behavior as the schema evolves.

3. Property roles

Role	Meaning	Examples
Parameter	Persistent model/baseline quantity.	Optimal fiber length; PCSA; pennation; maximum isometric force; tendon slack length.
Function of state	Computed from configuration rather than represented by one universal scalar.	Moment arm $r(q)$; musculotendon length $LMT(q)$.
State / output	Changes during simulation or is produced from the current state.	Muscle force $F(t)$; activation $a(t)$; joint angle $q(t)$.

This prevents a state-dependent quantity from being mistaken for an intrinsic anatomical constant. It also aligns naturally with muscle models that distinguish fixed musculotendon parameters from dynamic states used to calculate force.

4. Evidence and provenance

Every populated datum records the claim being supported, whether the value was directly reported, relevant source location, transformations, method, applicability, uncertainty, limitations, and review status.

```
property_id: optimal_fiber_length
value: 0.1569
unit: m
datum_kind: literature_derived
uncertainty: SD = 0.0257 m
source_claim:
  source_id: SRC:WARD_2009
  supports: reported_fiber_length
  directly_reported: true
transformation:
  15.69 cm -> 0.1569 m
applicability:
  level: cohort
limitations:
  - cadaver-derived
  - advanced-age cohort
```

5. Gluteus maximus: the reference-muscle template

Ward et al. provide experimental lower-extremity muscle architecture, while Arnold et al. demonstrate how such architecture can be translated into a Hill-type lower-limb model. Keeping those layers distinct gives us an experimental evidence layer and a model-translation layer rather than a single opaque parameter source.

Muscle mass	Literature-derived observation
Retain the reported cohort value and uncertainty; convert units transparently.	
Muscle length	Literature-derived observation
Keep the source definition/protocol and cohort applicability attached.	
Optimal fiber length	Literature-derived + model context
Ward supplies architecture; Arnold supplies model-translation context.	
Pennation angle	Literature-derived + model context
Preserve the reported architecture separately from its use in a muscle model.	
PCSA	Literature-derived architecture
Keep measurement/calculation method, cohort, and uncertainty with the value.	
Maximum isometric force	Model-derived
Store the PCSA input, specific-tension assumption, equation, and final force as a derivation graph.	
Tendon slack length	Model-dependent
Leave unresolved unless a defensible model-specific value and calibration context are supplied.	
Moment arm	Function of state
Represent as $r(q)$; interpretation depends on geometry, coordinate, and joint configuration.	
Muscle force	State/output
Compute during simulation from activation and musculotendon state; do not store as a fixed anatomical constant.	

6. Why the derivation graph matters

A final simulation value can be several steps removed from an experiment. For maximum isometric force, the model may begin with architecture such as PCSA, introduce a specific-tension assumption, apply normalization or scaling, and only then produce the force parameter. The final number should not erase that history.

Evidence chain:

Experimental architecture -> reported/derived PCSA -> model normalization or transformation -> specific-tension assumption -> maximum isometric force -> LB-AO baseline -> simulation -> future subject-specific override.

This lets us answer both *What number did the simulation use?* and *Why was that number justified?*

7. Bilateral symmetry as a reversible baseline assumption

For the initial generic model, one evidence-backed baseline initializes both left and right gluteus maximus. The two sides still remain independent ontology targets. A later measurement can override either side without changing the original evidence record or restructuring the model.

Shared baseline -> symmetric initialization -> independent LEFT/RIGHT parameter slots -> later personalization or asymmetry.

8. Unresolved values are informative

A null/unpopulated value is a documented knowledge gap, not a failure. It can later be resolved using stronger literature, an explicit model assumption, calibration, or subject-specific measurement. The validator should reject an incomplete datum as simulation-ready while clearly explaining what is missing.

9. Coordinate conventions

The ISB joint-coordinate recommendations provide a standardized basis for reporting hip motion. LB-QPS therefore requires coordinate context for quantities whose meaning depends on frame, axis, sign convention, or reference pose. This applies especially to joint angles, joint centers, forces, moments, attachment positions, and moment arms.

10. Relationship to OpenSim

OpenSim is treated as a computational modeling reference, not anatomical ground truth. Its muscle formulations and lower-limb models are valuable examples of how anatomical and experimental information becomes executable model parameters. LB-AO keeps the underlying evidence and transformations visible so an OpenSim-compatible parameter can still be traced back to its scientific basis.

11. Backward compatibility

Scientific traceability and software stability are linked. Stable identifiers should remain stable when meanings do not change; schema revisions should inherit or migrate older data where practical; the complete accumulated test suite should run on every release; and intentional breaking changes should be versioned with a migration path.

12. Why this is a useful foundation for the digital twin

The schema supports a progression from a generic evidence-backed musculoskeletal model to a personalized digital twin without discarding the original scientific basis. Generic literature values can initialize a model; later measurements can override them; uncertainty can inform sensitivity analysis; and simulation outputs can be traced back to the parameters and assumptions that produced them.

The gluteus maximus is therefore not just the first populated muscle. It is the reference implementation for how future muscles should be represented: anatomy first, quantitative claims second, explicit transformations third, and simulation use only after validation.

References and source roles

Ward SR, Eng CM, Smallwood LH, Lieber RL. (2009). *Are Current Measurements of Lower Extremity Muscle Architecture Accurate?* Clinical Orthopaedics and Related Research, 467(4), 1074-1082. DOI: 10.1007/s11999-008-0594-8.

Role in LB-AO: Primary experimental architecture source. Used for lower-extremity architectural observations and for demonstrating why cohort, method, and uncertainty belong with each value.

Arnold EM, Ward SR, Lieber RL, Delp SL. (2010). *A Model of the Lower Limb for Analysis of Human Movement.* Annals of Biomedical Engineering, 38(2), 269-279. DOI: 10.1007/s10439-009-9852-5.

Role in LB-AO: Model-translation source. Demonstrates conversion of experimentally informed architecture into Hill-type lower-limb muscle-tendon representations.

Wu G, et al. (2002). *ISB recommendation on definitions of joint coordinate system ... Part I: ankle, hip, and spine.* Journal of Biomechanics, 35(4), 543-548. DOI: 10.1016/S0021-9290(01)00222-6.

Role in LB-AO: Biomechanics standard used to justify explicit coordinate frames and joint-motion conventions.

OpenSim documentation: Thelen muscle model and lower-limb model documentation.

Role in LB-AO: Computational implementation reference used to compare parameter/state distinctions and trace how literature-derived quantities enter executable musculoskeletal models.

Project standard. Before a quantitative item enters LB-AO, classify what kind of claim it is, attach evidence appropriate to that claim, retain transformations and uncertainty, state its applicability and assumptions, and validate it before simulation use.